

Article

Performance of Professional Soccer Players: Multilevel Classification Using Machine Learning and SHAP Explainability by Playing Position

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Abstract

Background. Recent advances in data systematization have enabled the development of machine learning models to evaluate performance in elite sports; however, studies are needed to analyze the performance of professional players in relation to their playing position. **Objective.** To analyze the performance of professional soccer players who competed between 2017 and 2024 by applying a multilevel classification approach that integrates different machine learning algorithms. **Materials and Methods.** We analyzed 9088 player-seasons from professional players during the 2017–2024 seasons. These data were extracted from standardized databases on sports performance analysis belonging to the following leagues: LaLiga (Spain), Premier League (England), Bundesliga (Germany), Serie A (Italy), and Ligue 1 (France). The final sample, distributed by performance level, was as follows: elite players ($n = 1818$; 20%), mid-level players ($n = 2727$; 30%), and low-level players ($n = 4543$; 50%). The average age of the players analyzed was 26.1 ± 4.0 years, distributed across three outfield playing positions: center backs (CB), central midfielders (CM), and strikers (ST); goalkeepers were excluded because the dataset contains no goalkeeper-specific performance metrics. **Results.** Under a leakage-controlled protocol (the label-defining indicators were excluded from the predictors and the train/test split preceded all preprocessing), a linear model (logistic regression) achieved the best overall performance (mean macro-F1 = 0.729), ahead of ensemble and kernel-based methods; this indicates that, once target leakage is removed, the classification does not require non-linear models. Strikers (ST) were the most separable position (best macro-F1 = 0.818, area under the receiver operating characteristic curve [AUC-ROC] = 0.948) and center backs (CB) the most difficult (macro-F1 = 0.602, AUC-ROC = 0.791), with midfielders (CM) intermediate (macro-F1 = 0.770, AUC-ROC = 0.916). **Conclusions.** The results confirmed that



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performance structures differ substantially depending on the position in the field, supporting the use of position-specific analytical strategies. In this context, the combination of position-stratified dimensionality reduction, handling of imbalance, and explainable artificial intelligence allowed for the identification of interpretable performance patterns associated with the profiles of elite, mid-level, and low-level players.

Keywords: machine learning; SHAP explainability; player performance classification; class imbalance; performance analysis; SMOTE; logistic regression; sports analytics

1. Introduction

Soccer, as one of the most widely played and analyzed sports worldwide, has sparked growing interest in research aimed at understanding and optimizing the processes associated with athletic performance. In this context, advances in the longitudinal and systematic collection of data across different leagues, coupled with the development of machine learning (ML) models, have enabled a deeper analysis of performance from a multifactorial perspective. This evolution responds to the need to address the inherent complexity of soccer performance, which is determined by the dynamic interaction of physical, technical, tactical, and contextual dimensions.

Traditionally, performance analysis has relied on descriptive indicators, including physical parameters such as distance covered, as well as technical game variables [1,2]. However, the exponential growth in data availability has driven the transition toward more advanced metrics, based on multivariate approaches that seek to explain and predict complex phenomena such as injury risk, talent identification, and individual or team performance [3–5]. Consequently, it is necessary to implement more robust analytical strategies that enable the transformation of large volumes of information into useful knowledge for training design, decision-making [6], and performance management [7].

In this context, machine learning has emerged as a key tool due to its ability to model nonlinear relationships, handle large datasets, and dynamically adapt to new information. Its application in soccer has spanned various areas, such as predicting match outcomes, estimating player value, ranking teams in leagues, and analyzing individual performance [8,9]. To this end, multiple algorithms have been used, notably Random Forest, Decision Tree, Logistic Regression, Support Vector Machine, Gaussian Naïve Bayes, Multi-Layer Perceptron, and K-Nearest Neighbor (KNN); comparing these allows for the identification of the most suitable models based on the nature of the problem and the available data.

Recent studies have demonstrated the potential of machine learning when data balancing techniques and explainability methods are integrated. In this regard, Xu et al. [10] demonstrated that the combined use of algorithms such as Logistic Regression, Decision Tree, and Random Forest, along with balancing techniques like the Synthetic Minority Over-sampling Technique (SMOTE), yields interpretable and useful models for injury prevention, further relying on explainability methods such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to identify key predictors and assess individualized risk. Similarly, Qian and Lee [11] developed a predictive model for overtraining in elite youth soccer players using Random Forest with SMOTE, successfully anticipating performance decline with high reliability up to one week before it occurred. Furthermore, in the field of outcome prediction, Atta Mills et al. [12] pointed out that combining different algorithms with normalization and sampling techniques such as SMOTE significantly improves model accuracy, highlighting the superiority of feedforward neural

networks in outcome prediction and the consistent performance of models like Random Forest and XGBoost across multiple predictive tasks.

In particular, the integration of machine learning with positional analysis has allowed for a deeper understanding of performance specific to a player's role on the field. Models such as XGBoost have proven useful in predicting talent selection based on physical, physiological, and technical-tactical variables [5], while the use of big data and neural networks has facilitated the identification of offensive and defensive tactical patterns associated with different playing positions, contributing to improved decision-making and technical-tactical control [13].

Despite these advances, significant challenges remain in the analysis of performance in professional soccer, particularly regarding the management of imbalanced data, the integration of multiple levels of analysis, and the need to interpret models in a way that is understandable for their application in real-world contexts. In this regard, the combination of data balancing techniques such as SMOTE with explainability methods such as SHAP represents an opportunity to improve both the accuracy and interpretability of machine learning models, facilitating their transfer to applied settings.

In this context, the present study aims to analyze the performance of professional soccer players during the 2017–2024 period using a multilevel classification approach that integrates various machine learning algorithms. Specifically, various predictive models are compared, class imbalances are addressed using the SMOTE technique, and the SHAP method is employed to explain and interpret the contribution of the variables, with an emphasis on the analysis of technical and tactical elements differentiated by playing position. This approach seeks to provide methodological and applied evidence that contributes to a more precise, interpretable, and contextualized understanding of performance in professional soccer.

2. Materials and Methods

2.1. Study Design

A retrospective study [14] based on the secondary analysis of a publicly available, open-access dataset analyzed 9088 player-seasons from professional players, restricted to the 2017–2024 seasons within a larger open-access dataset. The dataset ("Football Players 1992–2025 Top 5 Leagues + 2025–2026") was retrieved from the Kaggle open-data platform [15] and is built upon publicly accessible aggregated season-level performance statistics corresponding to the following leagues: LaLiga (Spain), Premier League (England), Bundesliga (Germany), Serie A (Italy), and Ligue 1 (France). Each record represents a player's performance over an entire season, including both domestic and international competitions. Because the dataset is fully anonymized at the level of season-aggregated technical performance indicators, contains no protected health information, and was obtained from open public records that do not require additional consent for secondary scientific use, the present study did not involve any direct interaction with human participants. All analyses were nonetheless conducted in accordance with the ethical principles of the Declaration of Helsinki [16] concerning privacy, data protection, and responsible use of secondary information sources.

2.2. Participants

The selection of the analyzed data was based on the following inclusion criteria: (1) playing at least 15 matches per season; (2) belonging to the 2017–2024 seasons, the period for which advanced performance metrics were recorded (before 2017 these metrics are structurally absent in the source); (3) having the playing position recorded in the database. The exclusion criteria were: (1) data on players who were recovering from a long-

term injury; (2) players with more than 80% of performance variables showing zero values. The final sample, distributed by performance level, was as follows: elite players ($n = 1818$; 20%), mid-level players ($n = 2727$; 30%), and low-level players ($n = 4543$; 50%). The average age of the players analyzed was 26.1 ± 4.0 years, distributed across three outfield playing positions: center backs (CB), central midfielders (CM), and strikers (ST); goalkeepers were excluded because the dataset contains no goalkeeper-specific performance metrics.

2.3. Definition of the Performance Level Variable

The performance level was derived from the score obtained through the analysis of the specific playing position, since this variable was not included in the original database. The following procedures were followed to define this variable: a weighted score was developed for each playing position based on the indicators with the greatest academic weight, as determined by the specialized literature and the principal investigator's expertise. For center backs, the variables were: tackles, interceptions, passing accuracy, and number of aerial duels won. For midfielders, the main variables were successful passes, passing accuracy, assists, and passes advancing toward the opponent's goal. For striker, the analyzed variables were related to goals, shots on goal, and assists. Thus, the normalization of scores was established on a 0–1 scale by normalizing the minimum and maximum values, thereby aiming to facilitate comparability across playing positions. Next, percentiles were defined for each position as follows: Elite (percentile > 80), Mid (percentile 50–80), and Low (<50). This classification is derived solely from the position-specific composite score and does not rely on any external player-rating system.

2.4. Definition of Variables

The source dataset contains 113 numeric columns, of which two are demographic (age and year of birth) and were not used. The remaining 111 performance, participation and team-context variables were extracted, and 6 further indicators normalized per 90 min of play (per-90) were derived, giving 119 variables in total. A complete list of all variables with their definitions and standard FBref/Opta abbreviations (e.g., Sh/90 = total shots per 90 min, SoT = shots on target, KP = key passes, PrgP = progressive passes, SCA = shot-creating actions, Tkl = tackles, Int = interceptions, Won = aerial duels won, Rec = passes received, Cmp% = pass completion percentage) is provided as Supplementary Material (Table S1), which also records the role of every variable in the analysis. This per-90 approach facilitates the comparison of players with varying levels of participation, eliminating the inherent bias resulting from differences in the number of minutes played by each player and their playing position.

These per-90 variables were identified through the analysis of key performance indicators, including goals, assists, progressive passes, tackles, interceptions, and aerial duels won. To this end, the formula used to calculate the variables was: $\text{variable_p90} = \text{cumulative_variable} / (\text{minutes_played} / 90 + \text{epsilon})$, where epsilon = 0.01 prevents division by zero.

Because the analysis was restricted to player-seasons with recorded advanced metrics (2017–2024), the analytic sample was complete: the analytic matrix contained no missing values ($9088 \text{ records} \times 97 \text{ columns} = 881,536 \text{ cells}$; 0% missingness), so no imputation was required. All infinite or extreme values resulting from per-90 divisions were replaced with zero. In addition, eight difference and plus/minus indicators (G–xG, np:G–xG, A–xAG, +/–, +/–90, On–Off, xG+/–, xG+/–90) are provided by the source dataset after a positive-part transformation, in which negative values are clamped to zero ($\max(x, 0)$); these columns were used as supplied, so their minimum equals 0.00 by construction rather than as a result of preprocessing performed in this study. Characteristics with zero

variance (constants) were automatically suppressed. Fourteen of the 111 source variables were excluded before modeling: eight UEFA Champions League columns, which are recorded only for the subset of players whose clubs qualified for that competition, and six honours columns, which register competition outcomes rather than on-pitch performance and would therefore introduce target leakage. The six derived per-90 indicators were retained for descriptive characterization and did not enter the predictor matrix. This left the analytic matrix of 97 variables reported above. For each playing position, the indicators used to construct that position's performance label were then removed from those 97 to prevent target leakage, giving 93 predictors for center backs (tackles, interceptions, pass completion percentage and aerial duels won removed), 93 for midfielders (completed passes, pass completion percentage, assists and progressive passes removed) and 94 for strikers (goals, shots on target and assists removed). Supplementary Table S1 records the role of each of the 119 variables, so that the predictor matrix used for every position can be reconstructed exactly.

2.5. Stratified Principal Component Analysis (PCA)

PCA was applied separately for each of the playing positions identified in the database. The position-stratified strategy allows for the identification of correlation patterns among performance variables that are not related across different playing positions. In this sense, variables that covary within one position are orthogonal to those of another position; that is, they have no linear dependence on another variable. Therefore, using PCA in this manner avoids latent bias in performance variables specific to each playing position.

The following procedure was used to identify the principal components (PCs): (1) standardization of each variable to zero mean and unit variance (z-score); (2) exclusion of characteristics with zero variance for each playing position. Thus, the optimal number of PCs per playing position was defined by the total accumulated variance (85%) and with at least two PCs. The derived PC scores consolidated the input representation for the various classifiers. This model offers the advantage of promoting orthogonality among features within the same input while eliminating multicollinearity and simultaneously improving computational efficiency by reducing dimensionality [17,18]. Other advantages include reducing the high-frequency noise generated by individual variables with high weights and facilitating data visualization through two-dimensional biplots (Table 1).

Table 1. PC by playing position.

	N Comp	Cum Var %	PC1 %	PC2 %	PC3 %
CB	21	85.3	28.3	15.9	8.5
CM	21	85.4	33.5	13.7	7.2
ST	20	85.3	39.1	9.5	8.4

Note: CB = center backs; CM = central midfielders; Cum Var % = cumulative explained variance (%); N Comp = number of principal components retained; PC1 % = variance explained by the first principal component (%); PC2 % = variance explained by the second principal component (%); PC3 % = variance explained by the third principal component (%); ST = strikers.

The table above shows the number of retained PCs, as well as the cumulative variance explained and the contribution of the first three PCs. Similarly, the table shows the total variance explained by each PC according to playing position.

2.6. Class Balance: SMOTE

One of the main challenges in model identification is class imbalance. For the present analysis, different values were obtained (Low = 50.0%, Mid = 30.0%, Elite = 20.0%). To avoid these imbalances caused by minority values that could lead to bias when identifying

players, SMOTE [19] was employed. SMOTE interpolates linear points by the features with the nearest points; specifically, for each minority class, it randomly selects one of the five nearest points, which are calculated using Euclidean distance in the PC space, to provide a new relationship based on a random interpolation coefficient Alpha in the range [0, 1]. Thus, SMOTE was applied exclusively within the training folds of each cross-validation split, in the principal-component space, and was never applied to the validation or test partitions [20]. This is necessary to prevent data leakage and ensure that reported metrics accurately reflect the model's generalization ability on unseen data. SMOTE thus made it possible to balance the number of instances across the three classes to match the level of the most frequent class.

2.7. Classification Algorithms and Hyperparameter Optimization Process

Twelve supervised algorithms were defined and evaluated; these were selected to cover the entire methodological spectrum of classification. Thus, various methods were used, including ensemble methods (Random Forest, Extra Trees, Gradient Boosting, HistGradient Boosting, AdaBoost), support vector machines (SVM with a radial basis function [RBF] kernel), regularized linear models (Logistic Regression), decision trees, neural networks (MLP), and instance-based methods (KNN). Where possible, XGBoost and LightGBM were also included, as these are optimized implementations of gradient boosting that have been widely adopted in practice by other studies [21].

This hyperparameter optimization process was performed using GridSearchCV with internal cross-validation and a macro-averaged F1-score evaluation metric, which weighs the performance of each performance class independently of its frequency. SHAP-based interpretable machine learning has previously been applied in multi-factorial predictive settings [22]. This exhaustive search for the most important hyperparameters of each algorithm is performed based on the number of estimators, maximum depth, learning rate, regularization parameter, and hidden layer architecture. This external validation for each model was performed using StratifiedKFold with $k = 5$ folds, preserving the proportion of positions in each partition. The reported values are: F1-Score (unweighted average of the F1 per class), F1-weighted (weighted by class frequency), accuracy, multiclass area under the receiver operating characteristic curve (AUC-ROC; One-vs-Rest, macro average), Matthews' Correlation Coefficient (MCC), and Cohen's Kappa. The MCC and Cohen's Kappa are particularly well-suited for different types of imbalanced data because they consider both true positives and true negatives, thereby ensuring a more robust measure of agreement between the prediction and the true label.

2.8. Explainability Using SHAP

To enhance the explainability of each model, we used SHAP, a method grounded in cooperative game theory that assigns each identified feature a marginal predictive contribution for each individual instance [23]. These SHAP values consider desirable mathematical properties based on efficiency (the sum of SHAP values required to equalize the difference between the model's prediction and the baseline value), as well as symmetry, additivity, and the dummy property.

For tree-based models (Random Forest, Extra Trees, Gradient Boosting, HistGradient Boosting, XGBoost, LightGBM), TreeExplainer [24] was used, which calculates the exact SHAP values by leveraging the tree structure with $O(T \cdot L \cdot D)$ complexity, where T is the number of trees, L is the number of leaves, and D is the maximum depth. For non-tree models (SVM, Logistic Regression, MLP, KNN, AdaBoost), KernelExplainer was used with a reference set of 100 randomly sampled instances.

Consequently, the SHAP values were calculated in the PC space (model inputs). To facilitate interpretation by sports professionals, these values were projected onto the original variable space using the PCA loading matrix: $\text{importance_orig} = \text{PCA_loadings} \times \text{SHAP_averages_PC}$. This procedure allows us to identify which actual performance indicators, e.g., passing accuracy, goals per 90 min—are the most decisive factors in classifying a player's level at their specific position.

2.9. Statistical Analysis

Each of the differences between performance levels (Elite, Mid, Low) was analyzed using the nonparametric Kruskal–Wallis test, which is suitable for making multiple comparisons with distributions that do not follow a normal distribution; because the performance levels were derived from a subset of these indicators, these level-wise comparisons are reported as descriptive characterizations of the performance profiles rather than as independent confirmatory hypothesis tests. Effect size is reported as eta-squared (η^2). Differences were considered statistically significant at a two-tailed p -value < 0.05 with Bonferroni correction for multiple comparisons. All analyses were performed using Python 3.10 [25] with the following libraries: scikit-learn $\geq 1.3.0$, imbalanced-learn ≥ 0.11 , shap ≥ 0.44 , numpy ≥ 1.24 , pandas ≥ 2.0 , scipy ≥ 1.10 , matplotlib ≥ 3.7 , seaborn ≥ 0.12 , xgboost ≥ 1.7 , and lightgbm ≥ 3.3 .

3. Results

3.1. Dataset

Figure 1 shows the distribution of players by position. The largest concentration of players is among center backs (CB, $n = 4261$), followed by central midfielders (CM, $n = 2980$) and striker (ST, $n = 1847$), which was the least represented position. The distribution across the three performance levels is shown in Figure 1B. The temporal evolution of the database is evident in Figure 1C, restricted to the 2017–2024 seasons, the period for which advanced performance metrics are available. The age distributions by performance level are shown in Figure 1D, revealing that elite-level players are concentrated in the 23–30 age range.

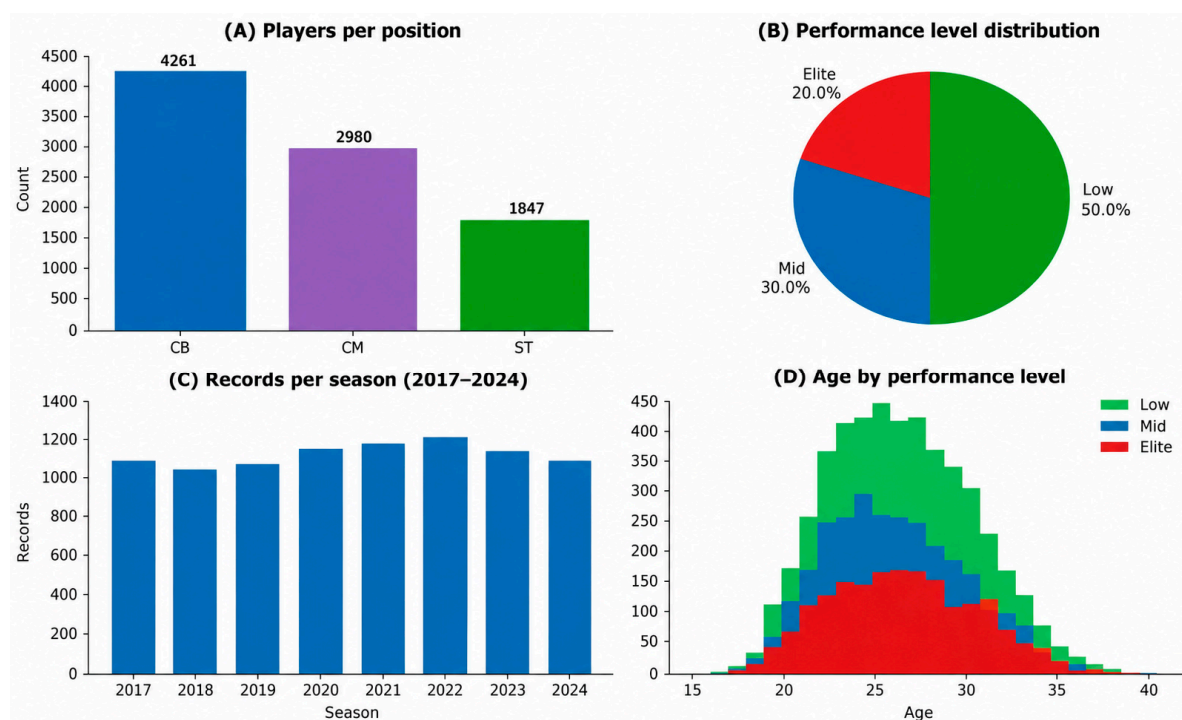


Figure 1. Overview of athletic performance in the dataset (2017–2024).

3.2. PCA and Playing Position

The stratification of the PCA results by playing position highlights the differences in player performance across different playing positions. Figure 2 shows the biplots (PC1 vs. PC2) for each playing position, with different colors representing the identified performance levels. The points represent a player identified by color, as follows: Elite (red), Mid (blue), and Low (green). Likewise, the separation of spaces between the three performance levels demonstrates the discriminability of the PCA space for subsequent classification. Finally, each figure represents a playing position; the first two PCs shown account for 44.2%, 47.2%, and 48.6% of the total variance for CB, CM, and ST, respectively, so the two-dimensional biplot captures only part of the structure (about 20 components are required to reach 85%; see Figure 3).

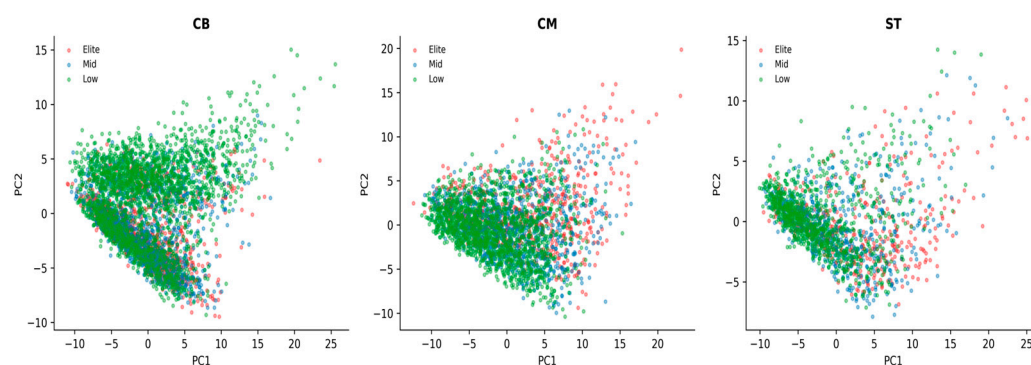


Figure 2. Biplots of the PCA (PC1 vs. PC2) by playing position. Note: PCA = principal component analysis; PC1 and PC2 = first and second principal components.

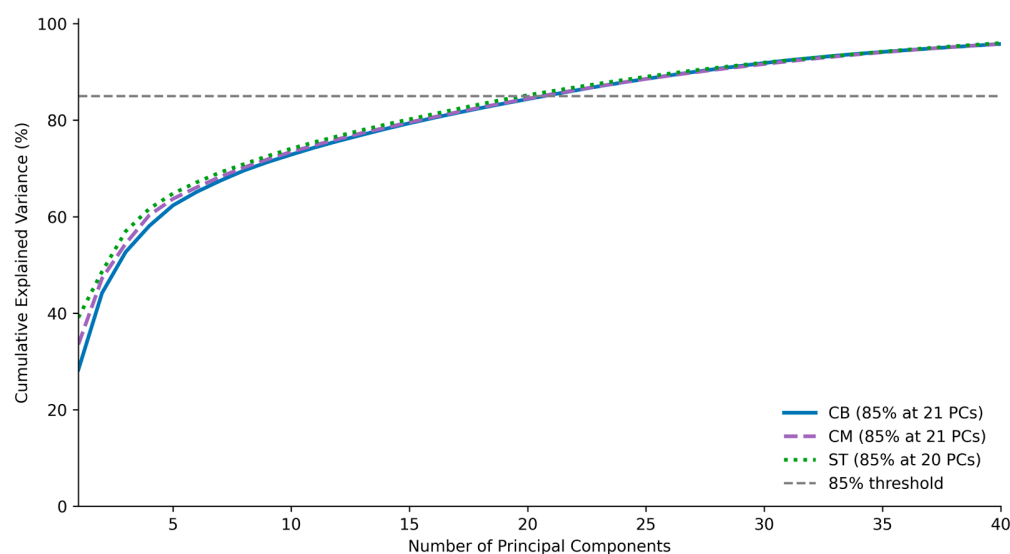


Figure 3. Cumulative explained variance as a function of the number of principal components for each playing position. Note: CB = center backs; CM = central midfielders; PC = principal component; ST = strikers.

In Figure 3, considering the cumulative variance explained by the number of PCs per playing position, it is evident that all three positions require a large number of components (about 20) to reach 85% of the variance, reflecting the multidimensionality of performance. Strikers (ST) concentrate slightly more variance in PC1 (39.1%), whereas center backs (CB) show the most gradual curve (PC1 = 28.3%). In contrast, center backs (CB) required the most gradual accumulation of variance to reach 85%, highlighting the multidimensionality of their performance profile, in which they must combine defensive efficiency (tackles,

interceptions) with offensive contribution (passing accuracy, progressive passes, expected goals [xG], key passes) and pressing ability (SCA, recoveries). This heterogeneity makes CB the most complex position to classify. The gray dashed line indicates the 85% cumulative variance threshold used as a criterion for selecting the appropriate number of components. Positions with steeper curves (ST) have a more one-dimensional performance, while positions with more gradual curves (CB) demonstrate greater multidimensional complexity.

3.3. Performance of Machine Learning Classifiers

Figure 4 shows the F1-Score heatmap for cross-validation ($k = 5$ folds) for the 12 classifiers evaluated for each of the three positions. This map allows for the immediate identification of performance patterns: the most intense values (yellow-red on the YlOrRd scale) correspond to the best classifiers, while lighter shades indicate lower performance. Thus, this scale allows us to identify which algorithms are more robust by playing position and where classification difficulties are concentrated. Under the leakage-controlled protocol, no single model family dominates the heatmap: a linear model (logistic regression) obtains the best average F1-score, ahead of the kernel-based (SVM) and ensemble models, indicating that once the label-defining variables are removed the classification does not require non-linear models. Striker (ST) shows the highest F1-scores (best macro-F1 = 0.818), indicating that the features distinguishing this position do so effectively, while center backs (CB) are the most complex to classify (macro-F1 = 0.602).

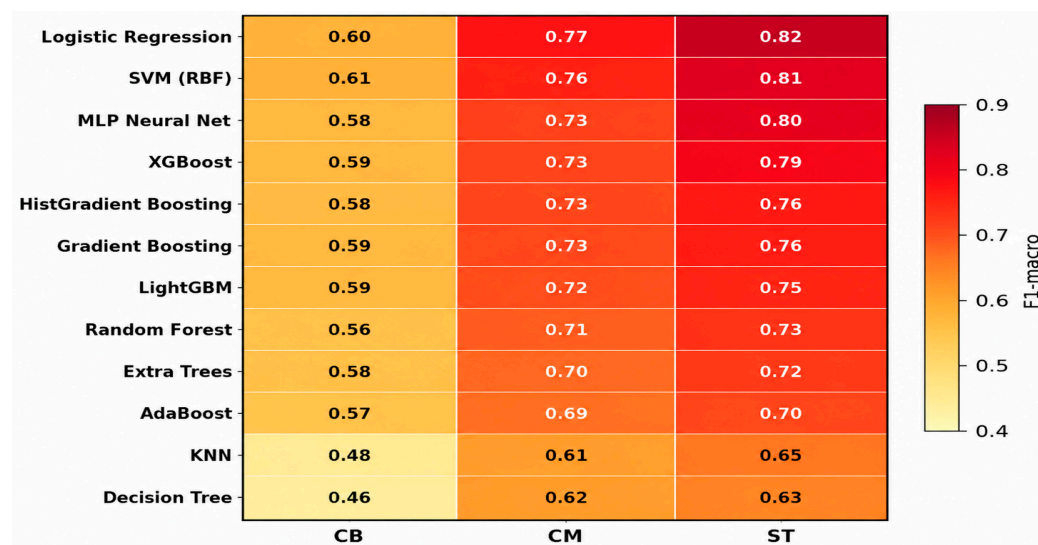


Figure 4. Heat map showing stratified cross-validation results for the classifiers evaluated by playing position. Note: AdaBoost = Adaptive Boosting; CB = center backs; CM = central midfielders; Extra Trees = Extremely Randomized Trees; HistGradient Boosting = Histogram-based Gradient Boosting; KNN = k-nearest neighbors; LightGBM = Light Gradient Boosting Machine; Logistic Regression = logistic regression; MLP = multilayer perceptron; RBF = radial basis function; ST = strikers; SVM = support vector machine; XGBoost = Extreme Gradient Boosting.

Figure 5 summarizes the categorization of ML classification algorithms, showing that logistic regression has the highest performance (F1-Score = 0.729) and the decision tree the lowest (F1-Score = 0.582). This indicates that the difference between the best and worst classifiers is 0.147 points, highlighting the need for algorithmic selection. The horizontal bars are ordered from best to worst performance overall. The classifier highlighted in red corresponds to the best model, blue to the second and third best, and gray to the rest. The numerical values above each bar indicate the exact average F1-Score.

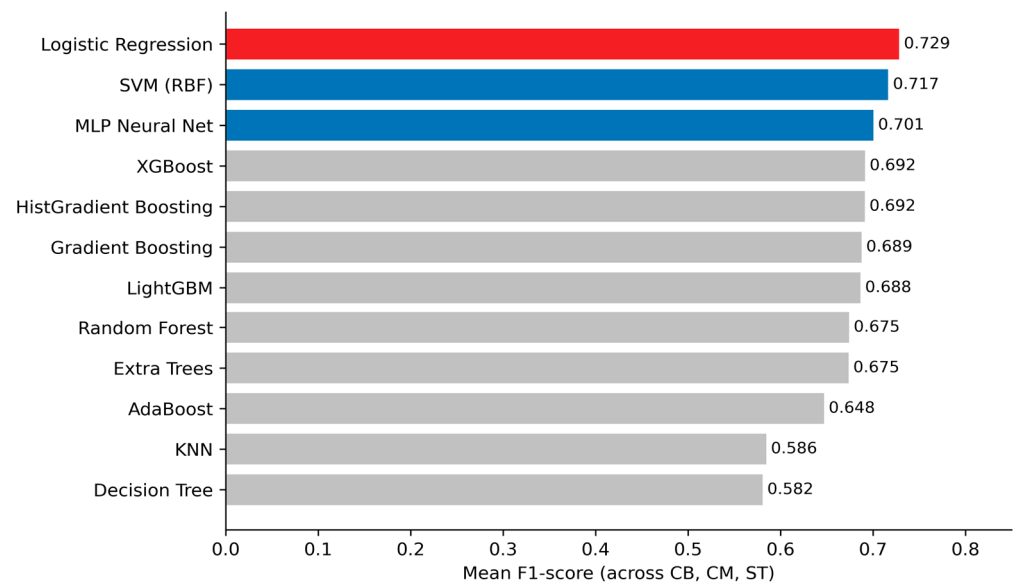


Figure 5. Overall ranking of classifiers based on the average F1-score for each of the three playing positions. Note: AdaBoost = Adaptive Boosting; CB = center backs; CM = central midfielders; KNN = k-nearest neighbors; LightGBM = Light Gradient Boosting Machine; MLP = multilayer perceptron; RBF = radial basis function; ST = strikers; SVM = support vector machine; XGBoost = Extreme Gradient Boosting.

Table 2 breaks down the performance of all classifiers across six depth metrics. It is evident that the values for Matthews' correlation coefficient (MCC) and Cohen's Kappa are consistently lower than those for accuracy, reflecting that even the best models make systematic errors in the least-represented class (Elite) prior to SMOTE balancing. This pattern retrospectively justifies the design of using SMOTE as a mandatory step before training.

Table 2. Overall Classifier Ranking.

	F1 Macro	Accuracy	MCC	Kappa	Precision	Recall
Logistic Regression	0.729	0.747	0.601	0.600	0.757	0.747
SVM (RBF)	0.717	0.736	0.583	0.581	0.748	0.736
MLP Neural Net	0.701	0.727	0.562	0.562	0.731	0.727
XGBoost	0.692	0.717	0.545	0.544	0.718	0.717
HistGradient Boosting	0.692	0.718	0.546	0.546	0.719	0.718
Gradient Boosting	0.689	0.711	0.542	0.540	0.719	0.711
LightGBM	0.688	0.713	0.539	0.539	0.714	0.713
Random Forest	0.675	0.704	0.525	0.524	0.701	0.704
Extra Trees	0.675	0.710	0.527	0.524	0.700	0.710
AdaBoost	0.648	0.657	0.469	0.462	0.688	0.657
KNN	0.586	0.597	0.392	0.385	0.645	0.597
Decision Tree	0.582	0.604	0.372	0.371	0.614	0.604

Note: F1 Macro = macro-averaged F1-score (harmonic mean of precision and recall); MCC = Matthews correlation coefficient; Kappa = Cohen's kappa coefficient; SVM = support vector machine; RBF = radial basis function; MLP = multilayer perceptron; KNN = k-nearest neighbors.

In Figure 6, each axis represents an evaluation metric (F1-score, accuracy, MCC, precision, recall). Thus, an ideal classifier would cover the entire pentagon. The shape of the radar plot reveals the balance between precision and sensitivity for each of the

algorithms used, allowing the most appropriate model to be selected based on the desired practical application.

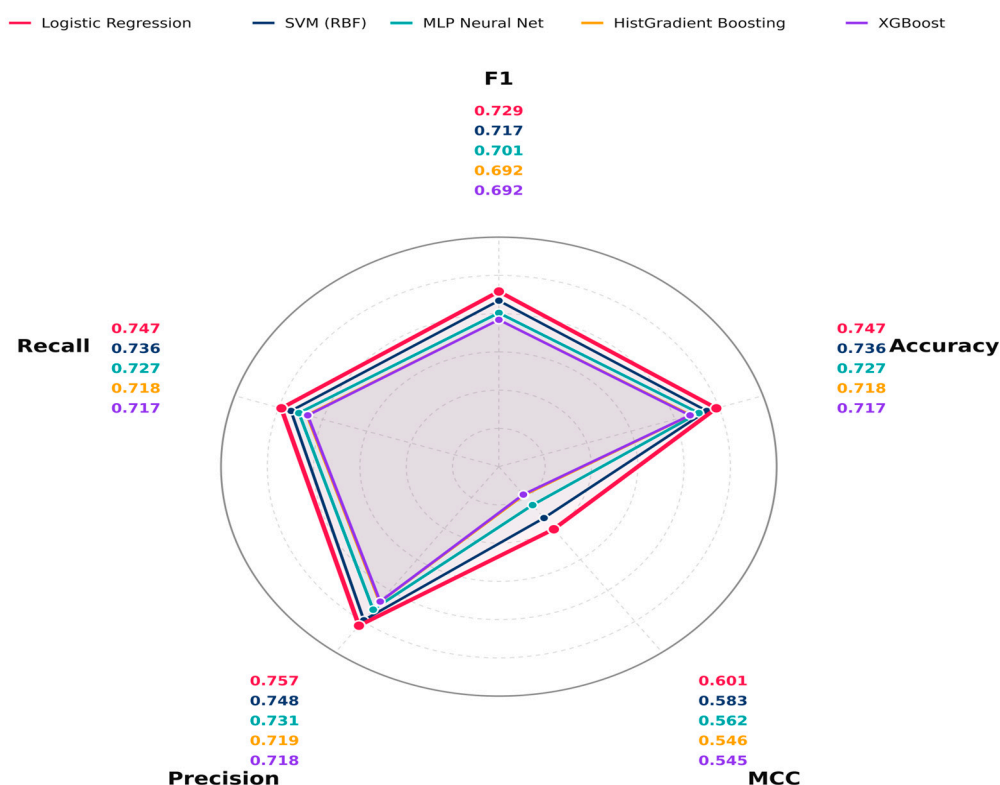


Figure 6. Multi-metric radar plot of the top five classifiers. Note: MCC = Matthews correlation coefficient; MLP = multilayer perceptron; RBF = radial basis function; SVM = support vector machine; XGBoost = Extreme Gradient Boosting.

Similarly, Table 3 shows the best classifier for each playing position based on cross-validation metrics. It shows that logistic regression (midfielders and striker) and SVM (center backs) were the most accurate for classifying performance levels by playing position, reflecting low standard deviations of the F1-Score across the five cross-validation folds (0.012–0.016; Table 3), which indicates high model stability. The AUC-ROC reached up to 0.95 for attacking positions, confirming stronger discriminative power there than for center backs (0.79). Each position indicates the selected algorithm along with cross-validation values (mean $\pm$ standard deviation of the F1-Score), as well as accuracy, MCC, Cohen's Kappa, and multi-class AUC-ROC.

Table 3. Best Classifier by Playing Position.

Playing Position	Classifiers	CV F1	CV F1 (SD)	Accuracy	MCC	Kappa	AUC-ROC
CB	SVM (RBF)	0.602	0.013	0.633	0.422	0.420	0.791
CM	Logistic Regression	0.770	0.016	0.782	0.652	0.651	0.916
ST	Logistic Regression	0.818	0.012	0.829	0.729	0.728	0.948

Note: CB = center backs; CM = midfielders; ST = strikers; SVM = support vector machine; RBF = radial basis function; CV F1 = cross-validated F1-score (harmonic mean of precision and recall); CV F1 SD = standard deviation of the cross-validated F1-score; MCC = Matthews correlation coefficient; AUC-ROC = area under the receiver operating characteristic curve.

3.4. Learning Curves and Generalization of Information

Figure 7 shows learning curves for the best classifier for each playing position (CB: SVM with RBF kernel; CM and ST: Logistic Regression), providing insight into generaliza-

tion performance. A persistent gap between the training and validation curves indicates overfitting (variance), whereas premature convergence suggests underfitting (bias). The figure shows the number of training samples on the x-axis; the blue line represents the training-set F1-score (mean $\pm$ standard deviation across folds), and the red shaded band represents cross-validation performance. Convergence of both curves as sample size increases indicates that the models generalize well and do not exhibit severe overfitting.

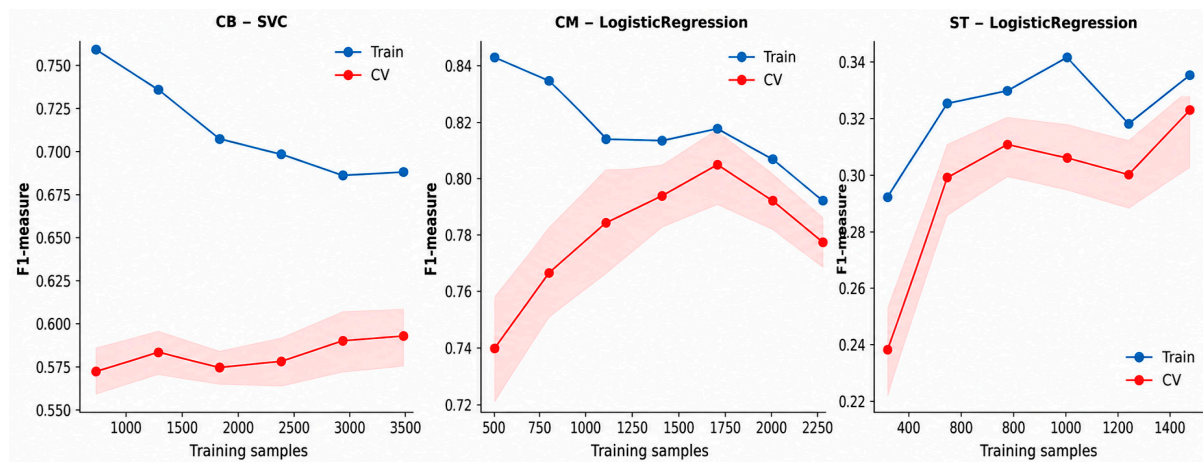


Figure 7. Learning curves for the best classifier for each playing position (CB: SVM with RBF kernel; CM and ST: Logistic Regression). Note: CB = center backs; CM = central midfielders; CV = cross-validation; F1 = F1-score; ST = strikers; SVM = support vector machine; RBF = radial basis function.

3.5. Confusion Matrices for Each Playing Position

The confusion matrices systematically display confusion patterns consistent with the structure of the performance analysis conducted. The most common errors stem from the discrimination between adjacent classes (Elite-Mid and Mid-Low). Thus, SMOTE helps introduce additional boundary instances to more rigorously evaluate the performance of the analyzed players. The Low category was identified as the most consistent across the three analyzed playing positions, indicating that the PCs effectively capture the main characteristics of performance. Meanwhile, the Elite class exhibits greater variability in PC accuracy, with higher classification rates for the ST and CM positions and lower rates for CB. In Figure 8, each row corresponds to a label (Elite, Mid, Low). The values assigned diagonally and shaded in dark blue represent correct classifications, while values outside the diagonal are identified as errors. This pattern and magnitude of errors reveal the confusion patterns that can be identified with the model to define immediate performance levels.

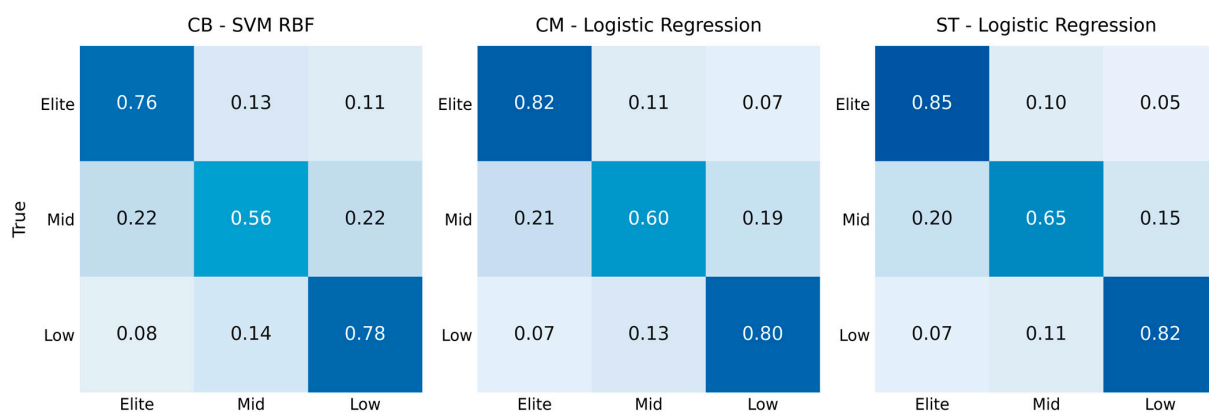


Figure 8. Confusion matrices for the best classifier for each of the playing positions analyzed. Note: CB = center backs; CM = midfielders; RBF = radial basis function; ST = strikers; SVM = support vector machine.

3.6. ROC Curves and Discriminatory Power

The Receiver Operating Characteristic (ROC) curves demonstrate the high discriminatory power of the trained models. The AUC-ROC exceeded 0.90 in most, but not all, position-level combinations: it did so consistently for CM and ST, whereas for center backs (CB) the Mid and Low levels remained below this threshold (AUC-ROC = 0.81 and 0.88, respectively), consistent with the lower aggregate AUC-ROC reported for this position in Table 3 (0.791), validating the overall quality of the classifiers beyond the F1-Score metric. As shown in Figure 9, the AUC-ROC (0.93) for the Elite category demonstrates the model's sensitivity and specificity in identifying players at this performance level, which has direct applications in talent scouting. Figure 9 shows the false positive rate (FPR) on the X-axis and the true positive rate (TPR) on the Y-axis. The diagonal line represents a random-order classifier (AUC-ROC: 0.50). AUC-ROC values above 0.90 indicate high discriminative power in most, though not all, position-level combinations.

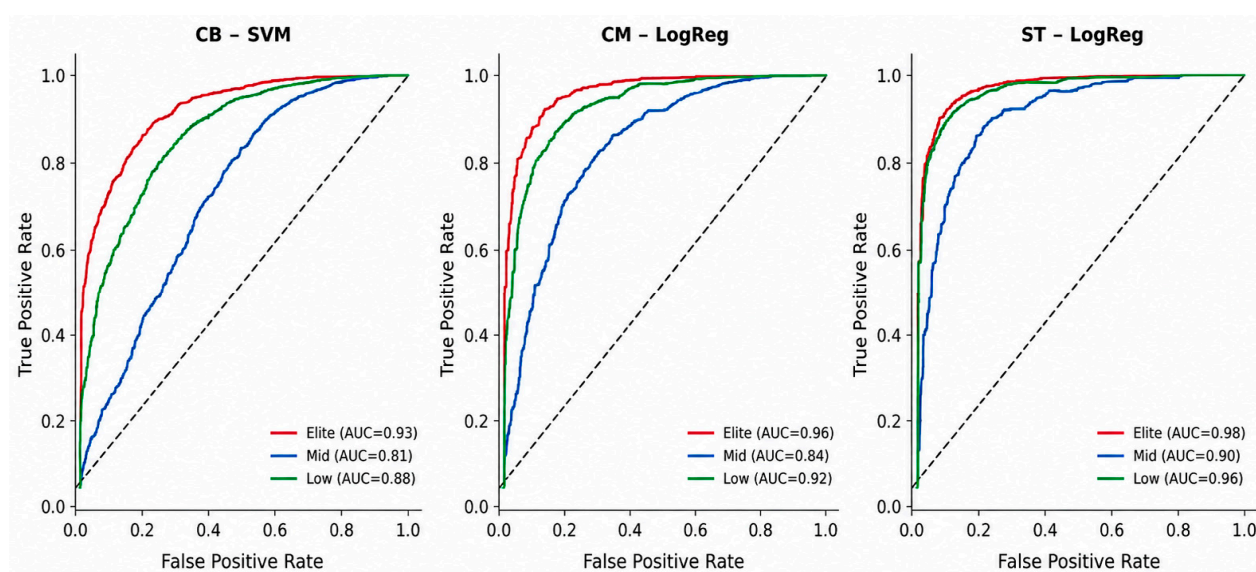


Figure 9. ROC curves by playing position, corresponding to the best model for each position. Note: CB = center backs; CM = midfielders; LogReg = logistic regression; RBF = radial basis function; ST = strikers; SVM = support vector machine; AUC = area under curve.

3.7. SHAP Explainability: Key Variables for Each Playing Position

The SHAP analysis is central to interpretability. Figure 10 shows the average SHAP importance (average SHAP value) for each PC according to playing position and the model used. This figure demonstrates how the model's results are translated into understandable language. Thus, the bars represent the average contribution of each PC in relation to the model's predictions. Longer bars indicate components that are more accurate in distinguishing performance levels (Elite, Mid, Low). This distribution indicates whether the model relies on critical dimensions or on a distributed representation.

For each playing position, the 12 variables that most influence the model's predictions are shown. This projection is calculated using $\text{importance_orig} = \text{PCA_loadings} * \text{average_SHAP_PCs}$. This identifies the performance indicators that serve as predictors for identifying performance according to each analyzed playing position. Figure 11 shows the SHAP results, which identify specific importance patterns by playing position using the 12 most influential variables.

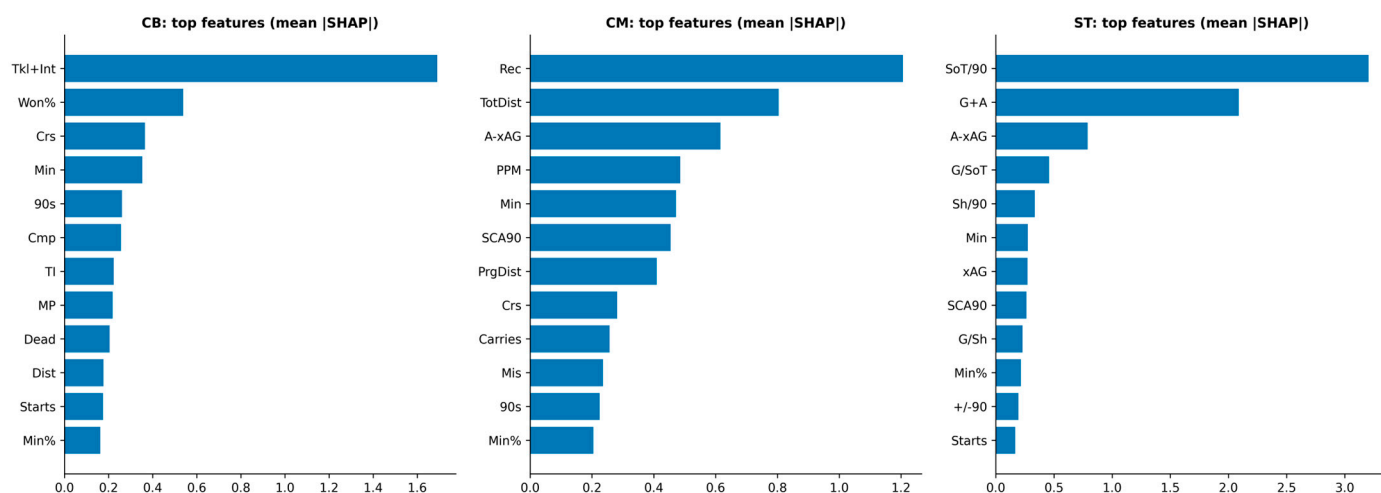


Figure 10. Average SHAP importance (average SHAP value) of principal components (PCs) for the best model for each position. Note: CB = center backs; CM = midfielders; ST = strikers; SHAP = SHapley Additive exPlanations.

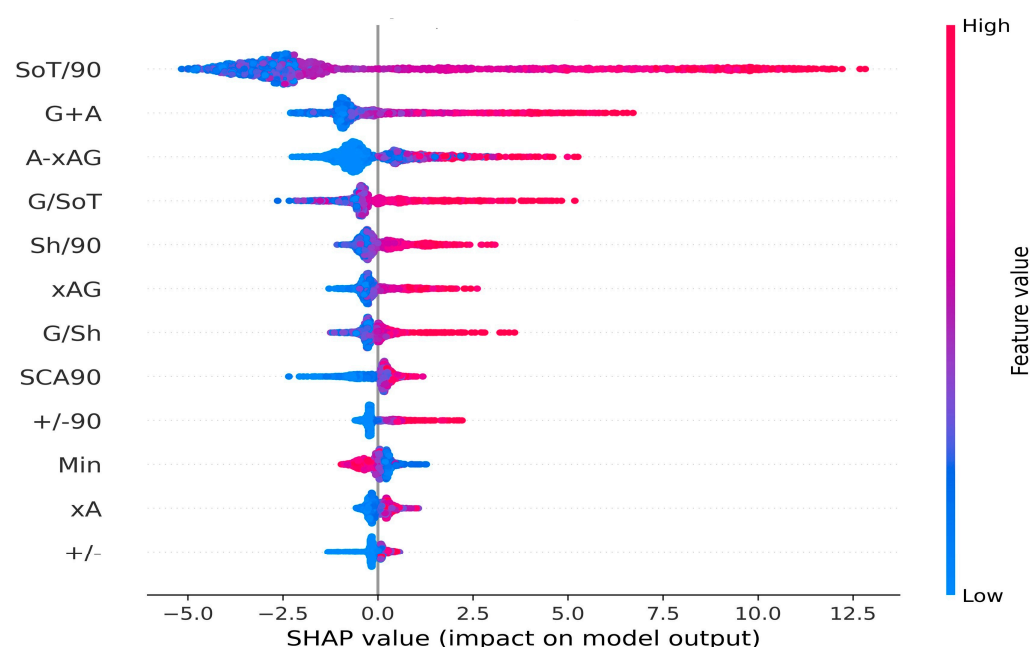


Figure 11. SHAP importance projected onto the space of original performance variables using the PCA loadings matrix. Note: PCA = principal component analysis; A-xAG = assists minus expected assisted goals; G+A = goals plus assists; G/Sh = goals per shot; G/SoT = goals per shot on target; Min = minutes played; SCA90 = shot-creating actions per 90 min; Sh/90 = shots per 90 min; SHAP = SHapley Additive exPlanations; SoT/90 = shots on target per 90 min; xA = expected assists; xAG = expected assisted goals; +/- = plus/minus; +/-90 = plus/minus per 90 min.

Figure 12 shows the SHAP analysis based on the Beeswarm plot, which displays the distribution of values for the most influential principal components. Each point represents an analyzed player (A), where the color indicates the value of the feature (red = high, blue = low). The horizontal dispersion shows the impact on the prediction. The SHAP dependency for PC1 shows how the value of this component influences the model's prediction (B).

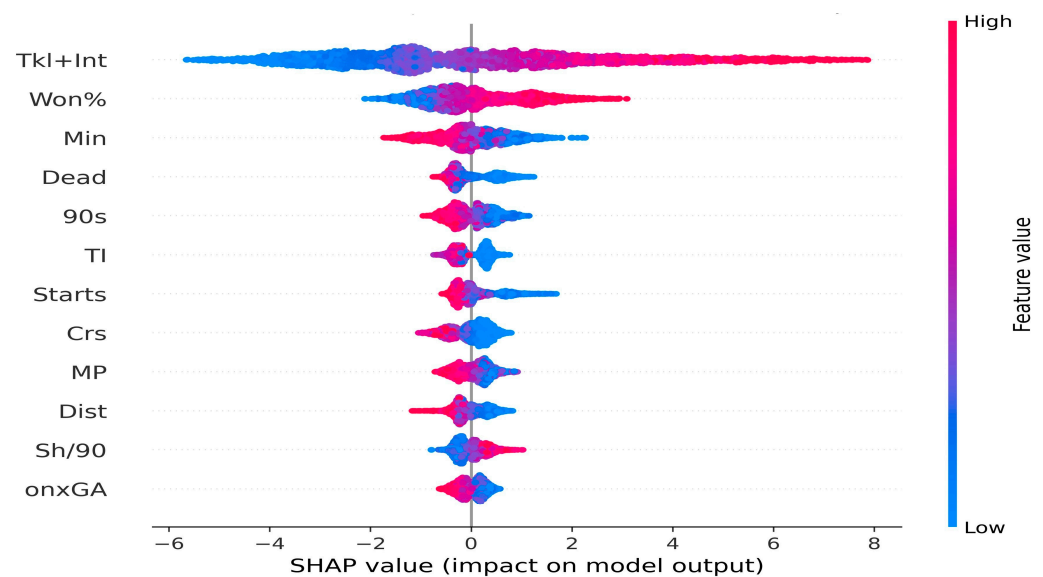


Figure 12. Detailed SHAP analysis for the position with the highest data availability. Note: SHAP = SHapley Additive exPlanations; 90s = full 90-min matches played; Crs = crosses; Dead = dead-ball situations; Dist = average shot distance; Min = minutes played; MP = matches played; onxGA = team expected goals against while the player was on the pitch; Sh/90 = shots per 90 min; TI = throw-ins; Tkl+Int = tackles plus interceptions; Won% = aerial duels won percentage.

Figure 13, meanwhile, shows the distribution of each PC by performance level (Elite, Mid, Low). Each bar illustrates the positive (red) or negative (blue) contribution of each PC (1 and 2) to the model's prediction for specific players. This allows a representative player to be classified within a specific performance level.

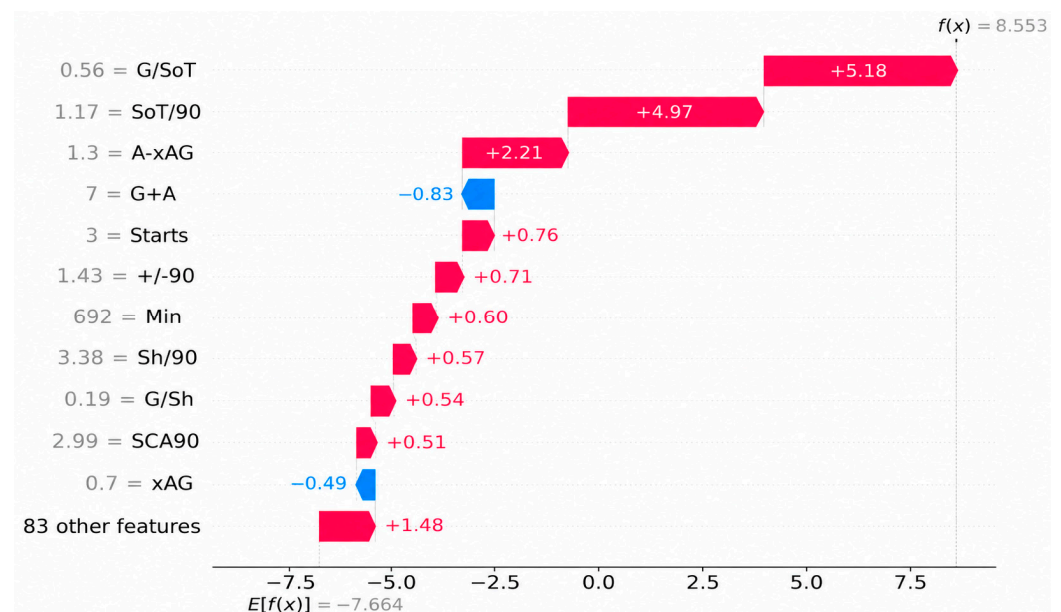


Figure 13. SHAP waterfall plots for representative players. Note: A-xAG = assists minus expected assisted goals; G+A = goals plus assists; G/Sh = goals per shot; G/SoT = goals per shot on target; Min = minutes played; SCA90 = shot-creating actions per 90 min; Sh/90 = shots per 90 min; SHAP = SHapley Additive exPlanations; SoT/90 = shots on target per 90 min; Starts = matches started; xA = expected assists; xAG = expected assisted goals; +/-90 = plus/minus per 90 min.

3.8. Performance Profiles by Position and Level

Figure 14 shows the profiles by playing position and performance level when considering 9 generalized dimensions, which were: (1) goals, (2) assists, (3) passing accuracy,

(4) tackles, (5) xG contribution, (6) progressive passes, (7) key passes, (8) aerial duels won, (9) crosses. Seven of these nine dimensions (excluding aerial duels won and crosses) were retained for the grouped-bar comparisons and Kruskal–Wallis tests below. The area covered by each polygon is also shown, illustrating how the distribution is proportional to the overall performance level across the dimensions represented for each playing position.

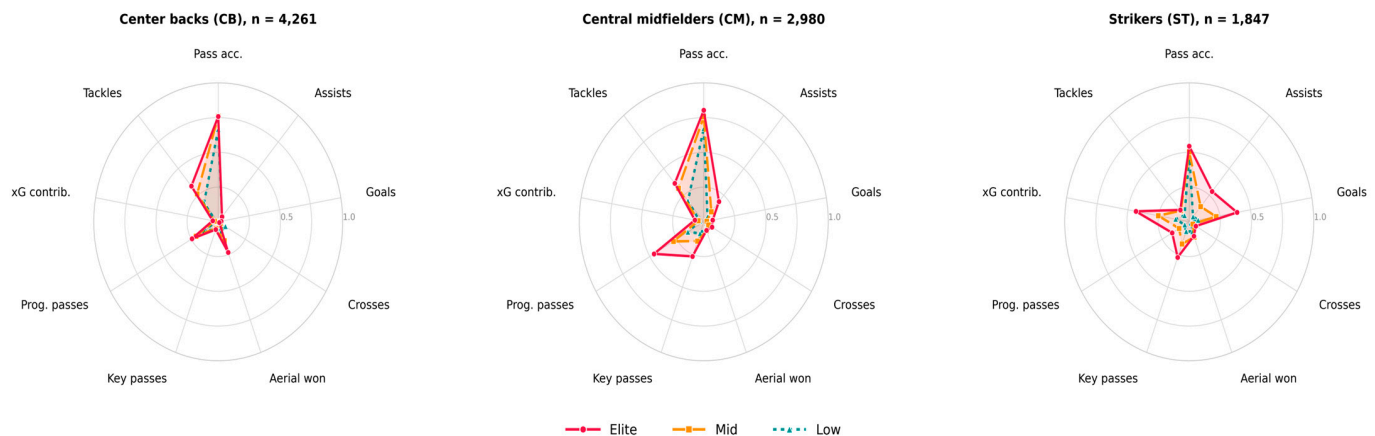


Figure 14. Radar charts show performance by playing position and performance level (Elite = crimson, Mid = amber, Low = teal) across 9 standardized dimensions. Note: each axis shows the position-wise median of the corresponding indicator per performance level, min–max standardized (0–1) to the pooled range of that dimension across all positions, so magnitudes are comparable across the three panels. xG = expected goals; CB = center backs; CM = central midfielders; ST = strikers.

The grouped bars allow for a direct comparison of the three performance levels for each playing position across the seven selected indicators: (1) goals, (2) assists, (3) passing accuracy, (4) tackles, (5) xG contribution, (6) progressive passes, (7) key passes. Differences across the three levels (Kruskal–Wallis) reached Bonferroni-corrected significance in 19 of the 21 position $\times$ indicator comparisons ($H = 51.0$ – 2048.6 , $df = 2$, adjusted $p < 0.001$; $\eta^2 = 0.027$ – 0.698). The two exceptions were assists ($H = 3.1$, $df = 2$, adjusted $p = 1.000$, $\eta^2 < 0.001$) and key passes ($H = 6.8$, $df = 2$, adjusted $p = 0.718$, $\eta^2 = 0.001$) in center backs, for which the three performance levels were statistically indistinguishable. This is consistent with the construction of the label, which for this position was derived from defensive and distribution indicators rather than from chance creation, and it illustrates that a player classified as elite within a position is not uniformly superior across every indicator. Because the levels were defined from a subset of these indicators, all of these comparisons are descriptive characterizations of the performance profiles rather than confirmatory hypothesis tests. Full statistics—medians and interquartile ranges by level, H , degrees of freedom, Bonferroni-adjusted p -values and eta-squared effect sizes—are reported in Supplementary Table S2. Figure 15 visualizes the corresponding position-by-level comparisons.

The boxplots in Figure 16 show the median, interquartile range, and outliers of the F1-Score for each of the algorithms. A robust classifier exhibits a narrow box and a high median, while high variability across positions indicates algorithm imbalance. The variability of the F1-Score for each of the three positions reveals that the classifiers show broadly overlapping F1-score distributions across positions, with the linear model (logistic regression) attaining among the highest medians once target leakage is removed; instance-based and simple tree models (KNN, decision tree) show greater variability, with excellent performance in some positions and poor performance in others, highlighting their sensitivity to the specific characteristics of each of the three positions analyzed in this study.

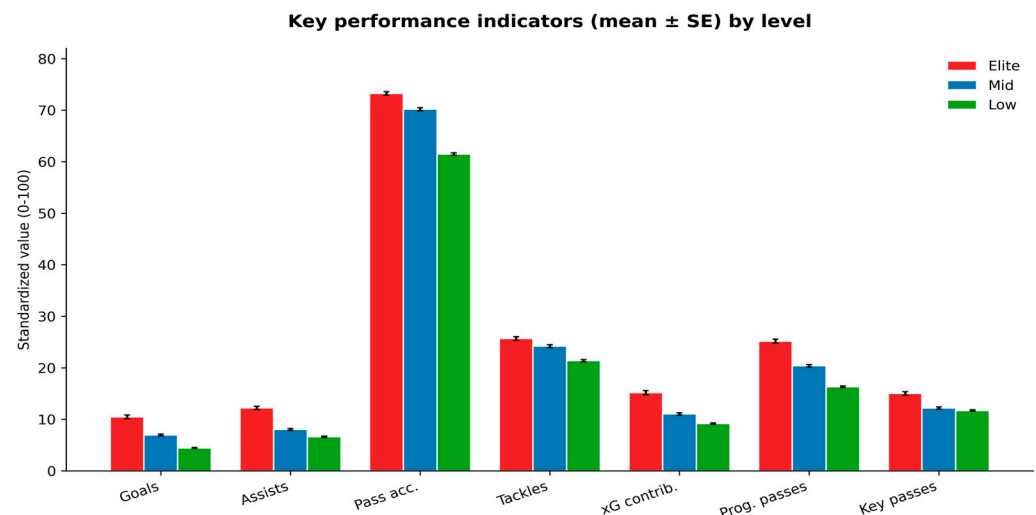


Figure 15. Key performance indicators by playing position and performance level. Note: CB = center backs; CM = central midfielders; ST = strikers; xG = expected goals.

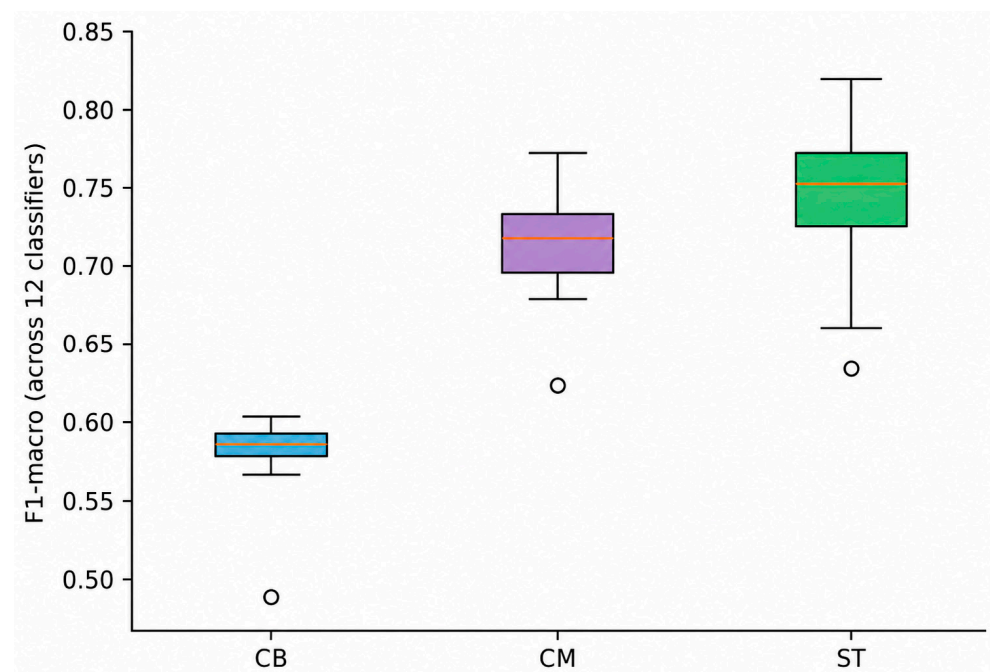


Figure 16. Distribution of the cross-validation F1-score by classifier across all active positions. Note: F1-score = harmonic mean of precision and recall; CB = center backs; CM = central midfielders; ST = strikers.

4. Discussion

The present study developed and evaluated an interpretable machine learning pipeline for the multilevel classification of professional soccer player performance across 9088 player-season records from 2017 to 2024. The analytical framework combined position-stratified principal component analysis (PCA), SMOTE-based class balancing, multiple supervised learning algorithms, and SHAP-based model explainability. Overall, the results indicate that player performance in professional soccer is better represented as a multifactorial, position-sensitive, and structurally complex phenomenon than as a set of isolated linear associations.

A central finding was that, under a leakage-controlled protocol, a linear model achieved the best overall performance. Logistic regression obtained the highest global macro-F1 (0.729), accuracy (0.747), and MCC (0.601), while SVM, the multilayer perceptron,

and the gradient-boosting ensembles (HistGradient Boosting, LightGBM, XGBoost) showed competitive but slightly lower performance; the decision tree produced the weakest global macro-F1 score. This pattern indicates that, once the label-defining variables are excluded and the train/test split precedes all preprocessing, the relationship between technical-tactical indicators and performance level is largely captured by linear decision boundaries. Although part of the sports analytics literature suggests that performance outcomes in team sports emerge from interacting, high-dimensional, and non-additive determinants, in the present leakage-controlled setting these interactions did not translate into an advantage for non-linear classifiers. Hecksteden et al. [26] argued that soccer-related outcomes should be understood as part of a non-linear “web of determinants”, particularly when multiple intrinsic, extrinsic, and time-varying factors interact. Similarly, Tuyls et al. [27] described soccer as a particularly challenging domain for artificial intelligence because player and team behaviors emerge from the interaction of statistical learning, computer vision, and game-theoretic decision-making. In contrast to this perspective, our results suggest that, after controlling for target leakage, models representing complex non-additive relationships did not outperform a well-regularized linear classifier for this position-specific classification task.

Gradient-boosting models nonetheless remained competitive and are theoretically well suited to structured tabular data, as they can capture interactions between predictors and are robust to heterogeneous feature distributions. For instance, Mănescu and Mănescu [28] used LightGBM as the primary classifier for team-sport athlete classification and reported strong discrimination, calibration, and interpretability when compared with logistic regression, Random Forest, and XGBoost. However, as their study was based on synthetic data, methodological support should be considered rather than direct empirical evidence for professional soccer. Additional support for gradient boosting approaches comes from Ming and Jin [29], who showed that Gradient Boosting Decision Trees (GBDT) were effective for identifying decisive performance factors in elite women’s volleyball, particularly because they captured non-linear associations between multidimensional technical indicators and match success. In soccer-specific research, Zareba et al. [30] also used XGBoost and deep learning approaches to model defensive player valuation, reinforcing the suitability of non-linear models for soccer-performance tasks involving context-sensitive and interaction-driven actions.

These findings also agree with broader reviews of soccer and sports analytics. Herold et al. [31] argued that machine learning offers an important advance over traditional notational analysis because it can identify patterns in large event and tracking datasets and support tactical understanding in professional soccer. Rico-González et al. [32] similarly concluded that machine learning applications in soccer have expanded across performance analysis, injury prediction, and talent identification, although future studies require careful attention to data quality, sample size, and predictive validity. Furthermore, Ghosh et al. placed sports analytics within a broader ecosystem that encompasses sensors, computer vision, wireless systems, deep learning, reinforcement learning, and statistical learning [33]. The current study contributes to this literature by applying an interpretable machine learning framework to large-scale soccer player-performance classification.

The position-stratified PCA represents one of the main methodological contributions of the study. Rather than reducing dimensionality globally, PCA was applied separately within each playing position, allowing the model to preserve role-specific covariance structures. This is important because the functional demands of center backs, central midfielders, and strikers are not equivalent. In the present analysis, each position required about twenty principal components to reach 85% of the variance, and the variance structure differed by position: CB showed the most gradual accumulation of variance (PC1 = 28.3%), whereas ST

concentrated more variance in the first component ($PC1 = 39.1\%$), with CM intermediate ($PC1 = 33.5\%$). These findings suggest that positional roles are characterized by distinct statistical structures in player-performance data. Tuyls et al. [27] noted that statistical learning methods, including PCA and player-vector representations, can help characterise player styles and roles. Similarly, Teixeira et al. demonstrated that AI-based tactical analysis in soccer increasingly relies on principal component analysis (PCA), clustering, t-SNE, neural networks, and graph metrics to represent tactical behavior, team dynamics, and movement patterns [34]. Therefore, the position-stratified PCA used in this study is not only a dimensionality-reduction step, but also a soccer-specific modeling strategy that respects the tactical heterogeneity of player roles.

The use of SMOTE was also justified by the class imbalance observed in the outcome variable. The low-performance class accounted for 50% of the sample, while the intermediate and elite classes accounted for 30% and 20%, respectively. Without appropriate balancing, models could have been biased towards the majority class and underperformed in detecting elite players. The application of SMOTE within the training folds in the PCA space was therefore methodologically important, as it reduced imbalance while limiting the risk of data leakage. This design is consistent with broader sports machine learning literature, where class imbalance is a recurrent issue, particularly in injury prediction, talent identification, and elite-performance classification tasks [26,28]. Davis et al. [35] also emphasized that evaluation in sports analytics requires careful handling of data dependencies, contextual factors, partitioning strategies, and the distinction between evaluating models and evaluating the indicators derived from them.

A second major contribution concerns explainability. The integration of SHAP enabled the translation of model outputs into interpretable performance indicators. This is particularly important in applied soccer settings, where models must support coaching, scouting, training, and recruitment decisions. SHAP analysis in the present study indicated that the most influential variables differed by position and were consistent with tactical expectations and domain knowledge. Furthermore, the projection of SHAP values from the PCA space back into the original feature space improved practical interpretability, allowing technical-tactical variables such as passing accuracy, xG contribution, progressive passes, tackles, and interceptions to be linked to model decisions.

This emphasis on interpretability is aligned with the growing literature on explainable artificial intelligence in sport. Shen [36] used SHAP to assess the sensitivity of machine learning models for soccer player market-value prediction, demonstrating how feature attribution can connect complex models with practical valuation decisions.

Mănescu and Mănescu [28] showed that SHAP can improve transparency in athlete selection models by identifying performance-critical predictors. Zaręba et al. [30] used advanced machine learning methods to improve the assessment of defensive actions, an aspect that is often underrepresented in traditional offensive metrics. Likewise, Davis et al. [35] further argued that performance indicators in sports must be intuitive, robust, trusted by domain experts, and evaluated beyond standard predictive metrics. The present study contributes to this movement by showing that explainability is not merely a technical add-on, but a necessary condition for translating machine learning into interpretable soccer-specific performance insights.

The findings also reinforce the need to conceptualize soccer performance as situationally influenced and tactically constrained. Previous studies have shown that the analysis of team sports requires, increasingly, the integration of event data, tracking data, tactical context, and patterns of interaction among players, as demonstrated by a study led by Settembre et al. [37] that examines how match outcomes in elite European soccer are influenced by contextual variables such as travel distance, Elo differences, match location,

recent performance, rest days, the coach's tenure, and squad rotation, although their unit of analysis was team-level match outcome rather than individual player classification, their findings reinforce the importance of contextualizing soccer performance rather than interpreting technical indicators in isolation.

Similarly, Teixeira et al. [34] described soccer as a complex, dynamic, and non-linear system in which tactical behavior emerges from the interaction between individual movements, collective organization, and situational constraints. Herold et al. also pointed out that machine learning can complement tactical knowledge by identifying attack patterns and interaction processes that traditional observational analysis might overlook [31]. In the present study, position-specific modeling and SHAP interpretation partly address this challenge by linking player-performance classification to role-specific technical-tactical structures.

4.1. Limitations

Several limitations should be acknowledged. First, the Elite/Mid/Low target variable was derived from a position-specific composite score rather than from an independent external label. Although this approach was necessary because the original dataset did not contain a validated performance-level outcome, it introduces construct-related uncertainty. In particular, because the composite label is computed from the same family of technical-tactical indicators that feed the models, the task is partly self-referential; although each position's label-defining variables were excluded from its own predictor set, residual information may remain distributed across correlated features. The reported classification metrics should therefore be read as the internal separability of the labeling structure rather than as prediction of an external ground truth. Future research should validate the classification against independent criteria such as expert ratings, national-team selection, external market valuation, awards, or established rating systems. This concern is particularly relevant because Shen [36] and other player-valuation studies show that market value is itself a multidimensional construct influenced by performance, age, reputation, contract status, and contextual variables. Therefore, external validation should not rely on a single criterion, but rather on convergent evidence across multiple independent indicators. In addition, because the source dataset contains no goalkeeper-specific performance metrics, goalkeepers were not analyzed; the present framework therefore applies to outfield positions (center backs, midfielders, and strikers) only.

Second, the present dataset did not include high-resolution tracking, Global Positioning System (GPS)/Electronic Performance and Tracking Systems (EPTS), or optical positional data. This limits the capacity to model off-ball movement, defensive pressure, spatial occupation, team compactness, and temporal tactical behaviors. Recent sports analytics reviews emphasize the growing importance of multimodal data, including wearable sensors, computer vision, optical tracking, GPS, and event-stream data, for improving real-time and context-aware performance analysis [33,34,38]. Herold et al. [31] similarly noted that tracking data enables more dynamic analysis of tactical behavior than event data alone. Incorporating such data would likely improve the contextual and tactical representativeness of future models.

Third, although cross-validation was used to estimate model performance, the study did not include a fully independent external validation across unseen leagues, competitions, or future seasons. This is an important limitation because soccer models may be sensitive to tactical culture, league style, data-provider conventions, and temporal changes in the game. In the context of sports medicine, Martin et al. [39] emphasized that external validation is essential for determining whether a machine learning model remains valid outside the development population. Although their study focused on anterior cruciate ligament

revision risk rather than soccer performance, the methodological principle is directly applicable: models intended for practical use must be tested on independent populations. Zaręba et al. [30] also noted that soccer models developed within one league may not generalize directly to other soccer contexts because of differences in tactical complexity, physicality, and competitiveness. Future work should therefore include temporal validation, cross-league validation, and external testing on unseen cohorts. Moreover, because soccer data availability is often greater in European competitions, the model may be affected by coverage bias, potentially limiting its applicability to players from South American, Asian, African, or lower-visibility leagues.

Fourth, although SHAP improves interpretability, feature attribution should not be interpreted causally. SHAP values identify predictive contributions within the fitted model; they do not prove that changing a variable would necessarily cause a player to move from one performance category to another. Therefore, the SHAP results should be understood as explanatory evidence of model behavior and as decision-support information, not as causal proof. This distinction is consistent with Davis et al. [35], who stressed the need to distinguish between evaluating the underlying predictive model and evaluating the usefulness, trustworthiness, and domain validity of the resulting sports indicator.

From an applied perspective, the present findings are relevant for scouting, talent profiling, training individualization, and market valuation. The pipeline can help identify players whose technical-tactical profiles resemble elite patterns within their position, detect development gaps in Mid-level players, and support evidence-based recruitment decisions. These applications are consistent with Tuyls et al. [27], who highlighted the growing role of AI in scouting, player evaluation, match preparation, and tactical analysis, and with Davis et al. [35], who organized sport machine learning applications around recruitment, match preparation, and player health management. However, practical use should remain complementary to expert judgment, tactical context, and longitudinal player monitoring.

Beyond club-level recruitment, the proposed framework may also be useful for identifying undervalued players in lower-visibility competitions, monitoring the development of young players over time, and supporting national-team talent mapping across heterogeneous league contexts [40], as well as in studies of women's soccer [41]. In addition, the SHAP-based outputs could be adapted for digital analytics platforms by providing transparent explanations of why a player is classified within a given performance level, thereby improving the communication of advanced performance metrics to coaches, analysts, scouts, and broader soccer audiences.

Future studies should expand the framework in four directions. First, external validation should be performed across leagues, seasons, and independent datasets. Second, tracking and contextual data should be incorporated to capture off-ball and tactical dimensions of performance. Third, longitudinal models should be developed to examine performance trajectories rather than isolated season-level classifications. Fourth, graph-based and relational machine learning methods should be explored to model player partnerships, team structure, defensive organization, and role-dependent interactions, building on recent work in player partnerships and defensive valuation [30,42].

4.2. Practical Applications

The study's main findings reveal several practical applications that could be useful for various stakeholders in the soccer ecosystem, including:

4.2.1. Scouting and Talent Identification

First and foremost, it facilitates intelligent scouting and talent identification processes through the automatic filtering of candidates by applying the classification model used

(Logistic Regression) to the various metrics of the players to be analyzed, determining the probability of them belonging to the Elite, Mid, or Low class based on the specific position required. It could also help identify undervalued players, especially those in less visible leagues, which could enhance market opportunities to “scout” players with high performance potential. Meanwhile, position-specific SHAP profiles could help select players with high scores on the identified SHAP variables that align with the tactical demands of the team’s playing system seeking to acquire their services. Finally, being able to track the progression of young players who show promise of high performance would foster a better understanding of how they develop by applying the model longitudinally to specific players or youth academies.

4.2.2. Individualized Training Planning

Identifying the SHAP values projected in the model could serve as a reference for physical trainers and technical staff regarding potential areas for improvement needed to enhance individual and team performance. Furthermore, these level classifications could serve as a reference for establishing specific improvement processes based on the expected impact at each performance level. In turn, position-specific SHAP profiles would allow for the development of training programs tailored to the specific demands of each playing role; for example, center backs would prioritize tackles, interceptions, and aerial duels, while central midfielders would prioritize passing accuracy and progressive actions, given that both roles report high values in the SHAP identification. Finally, the PC score should be monitored from season to season to assess whether a player’s trajectory in the latent space is approaching the elite cluster, thereby facilitating objective and quantifiable feedback.

4.2.3. National Teams and Long-Term Planning

National teams could use the pipeline to identify eligible players with talent across the various leagues in which they compete. The model used in this research allows for comparing the performance levels of players competing in different leagues by analyzing individual performance metrics beyond league-specific characteristics. Additionally, longitudinal analysis could provide insights into young talents to guide them toward the senior national team.

5. Conclusions

This study demonstrated that professional soccer player performance can be effectively classified using an interpretable and position-aware machine learning framework. Under a leakage-controlled protocol, a well-regularized linear model (logistic regression) achieved the strongest overall performance, ahead of ensemble and kernel-based approaches, indicating that non-linear modeling was not required once target leakage was removed.

The findings confirmed that performance structures differ substantially according to positional role, supporting the use of position-specific analytical strategies. In this context, the combination of position-stratified dimensionality reduction, imbalance handling, and explainable artificial intelligence enabled the identification of interpretable performance patterns associated with Elite, Mid, and Low player profiles.

The SHAP-based projections suggested that technical-tactical indicators were among the most influential features for performance discrimination relative to more general physical metrics, highlighting the importance of tactical efficiency and technical execution in modern professional soccer. Moreover, the coherence between SHAP-based feature attribution and expert tactical knowledge supports the practical interpretability of the proposed framework.

This study contributes a scalable, interpretable, and operationally meaningful framework for objective soccer-performance classification. From an applied perspective, the proposed approach may support scouting, recruitment, talent identification, training individualization, and performance profiling through transparent and data-driven evaluation processes.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/data11080190/s1>, The source dataset is publicly available at <https://www.kaggle.com/datasets/rodyanez/dataset-all-players-1992-2025> (accessed on 7 July 2026). Table S1: Complete list of the 119 variables considered, with their definitions, standard FBref/Opta abbreviations, role in the analysis and descriptive statistics; Table S2: Kruskal–Wallis comparisons of the seven key performance indicators across the three performance levels (Elite, Mid, Low) for each playing position, reporting the median and interquartile range per level, the H statistic, degrees of freedom, the Bonferroni-adjusted *p*-value and the eta-squared effect size.

Author Contributions: Conceptualization: B.A.B.-P. and J.F.L.-G.; Methodology: B.A.B.-P., R.V.-V. and R.Y.-S.; Software, data curation, and formal analysis: B.A.B.-P., D.A.R.-P. and J.D.P.-U.; Investigation: B.A.B.-P., W.G.V.-S. and R.V.-V.; Validation: J.F.L.-G., R.Y.-S. and R.V.-V.; Visualization: D.A.R.-P., J.D.P.-U. and W.G.V.-S.; Writing—original draft preparation: B.A.B.-P., R.V.-V., D.A.R.-P., J.D.P.-U. and W.G.V.-S.; Writing—review and editing: J.F.L.-G., R.Y.-S. and B.A.B.-P.; Supervision and project administration: J.F.L.-G. and R.Y.-S. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: This study is a retrospective, secondary analysis of an open-access, publicly available dataset, which compiles season-level aggregated technical and tactical performance indicators originally derived from public soccer statistical repositories. The dataset does not contain any private, sensitive, biometric, or identifiable health information, and no direct interaction with human participants was undertaken at any stage of the study. Because the research did not involve human subjects in the regulatory sense, did not collect new data, and exclusively used anonymized, publicly disseminated, non-clinical aggregated statistics, formal review and approval by an Institutional Review Board or Research Ethics Committee was not required under Spanish, Colombian, and Chilean national regulations governing research with secondary, non-identifiable, open-access data (i.e., Spain’s Ley Orgánica 3/2018 of Personal Data Protection, art. 27; Colombia’s Resolución 8430/1993 of the Ministry of Health, art. 11, classifying this type of study as “research without risk”; and Chile’s Ley 20.120, which restricts ethics-committee oversight to research involving identifiable human participants or biological samples). Consequently, the need for formal ethics approval was waived. The study was nonetheless designed and conducted in full accordance with the ethical principles of the Declaration of Helsinki [16], particularly with regard to scientific rigor, transparency, data integrity, and the responsible secondary use of publicly available information. As the research did not involve any direct interaction with human participants, informed consent to participate was not applicable.

Informed Consent Statement: Not applicable. The manuscript does not contain individual personal data, images, videos, or any identifiable information from human participants.

Data Availability Statement: The datasets analyzed during the current study are publicly available in the Kaggle repository under the title “Football Players 1992–2025 Top 5 Leagues + 2025–2026” [15], accessible at <https://www.kaggle.com/datasets/rodyanez/dataset-all-players-1992-2025> (accessed on 15 May 2026). The processed dataset, derived variables, and analysis code used in this study are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CB	Center Back
CM	Central Midfielder
CV	Cross-Validation
EPTS	Electronic Performance and Tracking Systems
FPR	False Positive Rate
GBDT	Gradient Boosting Decision Trees
GPS	Global Positioning System
KNN	K-Nearest Neighbors
LightGBM	Light Gradient-Boosting Machine
LIME	Local Interpretable Model-agnostic Explanations
MCC	Matthews' Correlation Coefficient
ML	Machine Learning
MLP	Multi-Layer Perceptron
PC	Principal Component
PCA	Principal Component Analysis
RBF	Radial Basis Function
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
ST	Striker (Forward)
SVM	Support Vector Machine
TPR	True Positive Rate
XGBoost	eXtreme Gradient Boosting
xG	Expected Goals.

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